

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
ICT 505 (DATA COMMUNICATION) TEST 1st SEMESTER (2023/2024) ANSWER ALL
QUESTIONS Time: 1 Hr

- 1 (a) Explain the OSI reference model in detail
(b) What are the advantages and disadvantages of TCP and UDP?
- 2 Given the dataword 100100 and divisor 1101 (Key)
(a) Show the generation of the codeword at the sender site (using binary division) via Cyclic Redundancy Check.
(b) Show the checking of the codeword at the receiver site (assuming no error)
- 3

For a (6,3) code, where n is 6 and k is 3 the generator matrix G is $G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$

- Find (i) All corresponding code vectors (ii) Minimum hamming distance " $d_{\min}$ "
(iii) Error detection and correction capability (iv) Parity check matrix
(v) Find error if received code is (100011)



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FIRST SEMESTER EXAMINATION (2023/2024) Session
ICT 505: (Data Communication and Network) Unit: 3

Instruction: ANSWER any four (4) questions in all

Time: 2 Hours 30 mins

- 1 (a) A DMS has the following alphabets with probability of occurrence as shown below:

Symbol	S ₀	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆
Probability	0.125	0.0625	0.25	0.0625	0.125	0.125	0.25

- (i) Generate the Huffman code for the source [4 marks]
(ii) Draw a table that shows the probability, digits obtained by tracing, codeword and number of bits per symbol for each symbol [4 marks]
(b) With the aid of a labelled diagram, explain the communication process in a simple data communication system [3 marks]
(c) List and explain the five components of a data communication system [4 marks]

process
sender
receiver
medium
protocol

- (a) The parity check matrix of a particular (7,4) linear block code is given by

[H] =

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$H = [P^T : I_{n-k}]$$

$$G = [I_k : P]$$

0.25
0.25
0.125
0.125
0.125
0.0625
0.0625
n = 7
k = 4

- (i) Find the generator matrix (G)

- (ii) List all the code vectors

- (iii) What is the minimum distance between code vectors?

- (iv) How many errors can be detected? How many errors can be corrected?

[2 marks]

[2 marks]

[2 marks]

[2 marks]

[4 marks]

- (b) Compare LAN, MAN and WAN

- (c) Give (5) benefits of computers in a networked environment when compared to computers in a standalone environment [3 marks]

- 3 (a) Find the checksum for the following bit sequence 10101001 00111001 and determine if there is any error at the receiver's end [6 marks]

- (b) Define the simplex, half-duplex and full-duplex modes of communication with examples [6 marks]

- (c) Give three advantages and disadvantages of the three modes of communication [3 marks]

if →

4. (a) Mr Cooper wants to send Ms Palmer an email. Mr Cooper composes his message in an email application on his laptop and he hits "send". Explain the communication process/flow of data between Mr Cooper and Ms Palmer using the OSI model [6 marks]
- (b) What are Network Protocols and how do they work? [4 marks]
- (c) Distinguish between TCP/IP model and OSI model [4 marks]
- (d) A sender sends 01110001; the receiver receives 01000001. If simple parity is used, can the receiver detect the error? [1 mark]
5. (a) An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows: The first group has 64 customers; each need 256 addresses, The second group has 128 customers; each need 128 addresses, The third group has 128 customers; each need 64 addresses. Design the subblocks and find out how many addresses are still available after these allocations [9 marks]
- (b) Why is Data Compression important? [3 marks]
- (c) State three advantages and disadvantages of Ethernet [3 marks]
6. (a) If the data unit of a generator polynomial is 100100 and the divisor is 1101. Determine the encoded data sent from the transmitter. What is the dividend at the transmitter? and show if the received data has an error or not using cyclic redundancy check [6 marks]
- (b) Give five differences between a Token Ring and Token Bus [3 marks]
- (c) Differentiate between a peer-peer network and a client-server network [6 marks]

Best of Luck

2 - 15
16 33
24 31



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B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: CPE505/ICT505/EEE501 **COURSE TITLE:** Data Communication and Networks. **UNITS:** 3 **TIME ALLOWED:** 150 minutes

Instruction: Answer any four (4) questions

Question One

- a. What is Encapsulation? (3 Marks)
- b. Explain the OSI model/encapsulation with aid of sequential diagram. (4 Marks)
- c. Write a short note on the following:
 - i. Cyclic redundancy check (CRC) (4 Marks)
 - ii. Forward error correction (FEC) (4 Marks)

Question Two

- a. Distinguish between asynchronous and synchronous communication. (6 Marks)
- b. Compare the OSI model and TCP/IP. (5 Marks)
- c. Differentiate between a LAN and a VLAN network with the aid of a diagram. (4 Marks)

Question Three

- a. What is P2P file sharing? (3 Marks)
- b. Describe Voice Over IP. (3 Marks)
- c. Write a short note on the following network security:
 - i. Authentication (3 Marks)
 - ii. Confidentiality (3 Marks)
 - iii. Integrity (3 Marks)

Question Four

- a. Explain Domain Name Server. (5 Marks)
- b. Write short note on the following switches connections:
 - i. Circuit switching (2 Marks)
 - ii. Message switching (2 Marks)
 - iii. Packet switching (2 Marks)
- c. Discuss Physical Network Security. (4 Marks)



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Question Five

- a. Describe the transport layer of TCP/IP. * (4 Marks)
- b. What is network Topology? Hence list and explain four types of Network Topology. u³ (5 Marks)
- c. A small business has a wireless network that consists of 6 devices. Describe the steps taken to assign an IP address in a computer network of the business workstation. (6 Marks)

Question Six

- a. Explain Denial of Service Attacks. (3 Marks)
 - b. Discuss the Spread Spectrum Wireless Transmission. (3 Marks)
 - c. What is Network Troubleshooting? (1 Mark)
- Hence, discuss (i) A Systematic Method for Troubleshooting; (ii) Procedures for Troubleshooting (8 Marks)